

Data Cleaning, Preprocessing and Feature Engineering

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Introduction

This project demonstrates data cleaning, preprocessing and feature engineering using two Kaggle datasets (Titanic and Housing). The goal is to improve data quality before applying machine learning algorithms.

Objectives

- Detect missing values
- Remove duplicate records
- Handle invalid values
- Encode categorical data
- Scale numerical features
- Create new features
- Split datasets
- Build reusable preprocessing workflow

Datasets

- **Dataset 1:** Titanic Dataset
- **Dataset 2:** Housing Dataset

Techniques Used

- Data Quality Assessment
- Missing Value Treatment
- Label Encoding / One-Hot Encoding
- Standardization / Min-Max Scaling / Robust Scaling
- Log Transformation & Feature Engineering
- Train/Validation/Test Split
- Pipeline and ColumnTransformer

Results

The project successfully cleaned both datasets. Missing values were handled, duplicates checked, numerical features processed, categorical variables encoded and cleaned datasets saved for future machine learning tasks.

Conclusion

The preprocessing pipeline prepared both datasets for machine learning. The project demonstrates essential preprocessing concepts in a beginner-friendly way.

Project Output Screenshots

Titanic Dataset Output

```
===== Titanic Dataset =====  
Shape : (891, 12)  
  
Missing Values  
PassengerId      0  
Survived          0  
Pclass           0  
Name             0  
Sex              0  
Age             177  
SibSp            0  
Parch            0  
Ticket           0  
Fare             0  
Cabin            687  
Embarked         2  
dtype: int64  
  
Duplicate Records : 0  
  
Data Types  
PassengerId      int64  
Survived          int64  
Pclass           int64  
Name             str  
Sex              str  
Age             float64  
SibSp            int64  
Parch            int64  
Ticket           str  
Fare             float64  
Cabin            str  
Embarked         str  
dtype: object  
  
Titanic Dataset Cleaned Successfully
```

Housing Dataset Output

```
===== Housing Dataset =====
```

```
Shape : (545, 13)
```

```
Missing Values
```

```
price          0
area           0
bedrooms       0
bathrooms      0
stories        0
mainroad       0
guestroom      0
basement       0
hotwaterheating 0
airconditioning 0
parking        0
prefarea       0
furnishingstatus 0
dtype: int64
```

```
Duplicate Records : 0
```

```
Data Types
```

```
price          int64
area           int64
bedrooms       int64
bathrooms      int64
stories        int64
mainroad       str
guestroom      str
basement       str
hotwaterheating str
airconditioning str
parking        int64
prefarea       str
furnishingstatus str
dtype: object
```

```
Housing Dataset Cleaned Successfully
```

Final Processing Output

```
ColumnTransformer Supported
Reusable Pipeline Created

Processing Titanic Dataset...
Mean / Median / Mode Supported
Advanced Imputation : KNN / Iterative (Concept)
Ordinal Encoding : Used for ordered categories
Rare Category Handling : Combine low-frequency values
High Cardinality : Reduce unique categories
Random Split Completed
Validation Split Completed
Stratified Split Completed
Time Split : Used only for date datasets
Avoid Data Leakage : Split before preprocessing
Titanic Dataset Saved

Processing Housing Dataset...
Mean / Median / Mode Supported
Advanced Imputation : KNN / Iterative (Concept)
Ordinal Encoding : Used for ordered categories
Rare Category Handling : Combine low-frequency values
High Cardinality : Reduce unique categories
Random Split Completed
Validation Split Completed
Time Split : Used only for date datasets
Avoid Data Leakage : Split before preprocessing
Housing Dataset Saved

===== PROJECT COMPLETED =====
```